

		Power loss = $I^2R = (15.2)^2(5) = 1100 \text{ W}$
29	A	$hf = \phi + E_{k,max}$ gives $\frac{hc}{\lambda} = \phi + E_{k,max}$, so $\frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{250 \times 10^{-9}} = 4.0 \times 10^{-19} + \frac{1}{2} \times 9.11 \times 10^{-31} \times v_{max}^2$ $v_{max} = 9.3 \times 10^5 \text{ m s}^{-1}$
30	D	$\ln 2 \quad \ln 2$